

HW # 22

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Problem 1:

$$\begin{aligned} |A| &= 1(3) + 1(-4-6) \quad \text{expanding } C_1 \\ &= 3 - 10 \\ &= -7 \end{aligned}$$

$$\begin{aligned} |B| &= 2(2+2) + 3(1) \quad \text{exp from } C_1 \\ &= 8 + 3 \\ &= 11 \end{aligned}$$

$$\begin{aligned} |AB| &= |A| |B| \\ &= -7(11) \\ &= -77 \end{aligned}$$

Problem 2:

$$\begin{aligned} |A| &= 10^3 \begin{vmatrix} 1 & -2 & 4 \\ 3 & 0 & 5 \\ 2 & -3 & 1 \end{vmatrix} \\ &= 1000 \left[-3(-2+12) - 5(-3+4) \right] \\ &= 1000 \left[-30 - 5 \right] \\ &= 1000(-35) \\ &= -35000 \end{aligned}$$

Problem 3:

$$\begin{aligned} |A| &= -2(-3-3) - 1(6+6) \\ &= 12 - 12 \\ &= 0 \end{aligned}$$

Inverse not possible matrix is singular.

$$\begin{aligned} |B| &= -2(3-3) - 1(6+6) \\ &= -12 \end{aligned}$$

Inverse exist matrix is non-singular.

Problem 4:

$$|A^{-1}| = 1/|A|$$

$$\begin{aligned} |A| &= 1(-2) + 3(2) \\ &= -2 + 6 \\ &= 4 \end{aligned}$$

$$|A^{-1}| = 1/4$$

Problem 5:

$$\begin{aligned} |A| &= -2(5-2) \\ &= -6 \end{aligned}$$

$$A^T = \begin{bmatrix} 3 & 2 & -4 \\ 1 & 0 & -1 \\ -2 & 0 & 5 \end{bmatrix}$$

$$\begin{aligned} |A^T| &= -2(5-2) \\ &= -6 \end{aligned}$$

$$|A| = |A^T|$$

Problem 6:

$$a) \begin{bmatrix} 0 & 2 & -1 \\ 3 & -2 & 1 \\ 3 & 2 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \\ -4 \end{bmatrix}$$

$A \quad X = B$

$$|A| = -2(-3-3) - 1(6+6) \\ = 0$$

Since $|A|=0$ there are infinitely many solutions.

$$b) \begin{bmatrix} 0 & 2 & -1 \\ 3 & -2 & 1 \\ 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \\ -4 \end{bmatrix}$$

$$|A| = -2(3+3) - 1(6+6) \\ = -12$$

Since $|A| \neq 0$ there is a unique solution.